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Low palaeolatitude of Late Proterozoic glaciation: early timing of remanence in haematite of the Elatina Formation, South Australia

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ABSTRACT

Previous palaeomagnetic study of a ~ 10 m thick rhythmite member of the Elatina Formation, part of the Late Proterozoic (~ 650 Ma) Marinoan glacial succession in South Australia, argued strongly for a low palaeolatitude (~ 5°) of deposition. However, a fold test was thwarted by the approximate alignment of the remanence with the axis of tectonic folding. Here we report results of a fold test on soft-sediment slump folds with a wavelength of 14–22 cm in the Elatina rhythmites, the axes of which are approximately perpendicular to the direction of remanence. Fifty-one standard palaeomagnetic specimens sampled across the slump structure gave $\alpha_{95} = 3.7°$ prior to unfolding, $\alpha_{95} = 1.8°$ with 67% unfolding, and $\alpha_{95} = 2.4°$ with 100% unfolding. The results indicate that the remanence was acquired very soon after deposition; apparently it was acquired prior to soft-sediment slumping and then slightly sheared by the disturbance. The Elatina pole must be considered a virtual geomagnetic pole because the tidal rhythmite member studied represents only 60–70 years of deposition. However, the mean pole position is similar to other Late Proterozoic poles for South Australia, which implies that the very low inclination (< 10°) for the Elatina rhythmites does not record a geomagnetic excursion or reversal but does indeed indicate deposition in low palaeolatitudes. The similarity of many of the Late Proterozoic pole positions is evidence also that inclination error is not significant. The low palaeolatitude of Late Proterozoic glaciation is one of the major enigmas in contemporary Earth science, raising questions concerning the nature of the geomagnetic field, climatic zonation, and the Earth's rotational parameters in Late Proterozoic time.

1. Introduction

Two major glaciations of Late Proterozoic age are recorded in the Adelaide Geosyncline in South Australia [1,2]. During the younger, the Marinoan Glaciation, at ~ 650 Ma glaciogenic sediments of the Elatina Formation and equivalents accumulated within the Geosyncline (Fig. 1) while a strongly seasonal periglacial environment prevailed on the Stuart Shelf to the west [3,4]. The stratigraphic relationship of the Elatina Formation to the principal Late Proterozoic units in the Adelaide Geosyncline is shown in Fig. 2.

Palaeomagnetic study of fine-grained clastic rhythmites from the Elatina Formation has provided perhaps the clearest evidence yet for a low palaeolatitude of Late Proterozoic glaciation [5]. The ~ 10 m thick rhythmite member of laminated siltstone and very fine-grained sandstone at Pichi Richi Pass (32°25′S, 137°59′E; see Fig. 1) displays conspicuous cyclicity that formerly was ascribed to solar-climatic variability [6]; however, the identification of tidal structures and periodicities in correlative rhythmites in South Australia and subsequently in the Elatina Formation itself indicates that the cyclicity is of tidal-climatic origin [7,8]. The Elatina rhythmite member evidently was deposited mainly from suspension below storm wave base on a distal ebb-tidal delta. Periodic changes in the thickness of semidiurnal and diurnal laminae, and of conspicuous fortnightly groups of such laminae (lamina-cycles), are taken to reflect variations in the amount of fine clastic material entrained and deposited by tidal currents in response to periodic changes in the velocity and range of palaeo-ebb tides. So interpreted, the

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member represents an unsurpassed palaeotidal record of about 60 yr duration that provides unique information on the Earth's palaeorotation and the past lunar orbit [7–10]. The glacial and periglacial affinity of the Elatina Formation has remained firmly established [1,4], and it is this glacial connection that makes palaeomagnetic study of the rhythmite member of special interest.

Previous palaeomagnetic work [5] demonstrated that the magnetisation of the rhythmite member of the Elatina Formation is extremely stable, typically showing appreciable decay only after heating to above 660°C. The magnetic carriers were identified as being mainly haematite and martite grains of high temperature (igneous or metamorphic) origin. It was concluded that since the Elatina Formation has undergone only burial diagenesis, the haematite is detrital and the remanence must

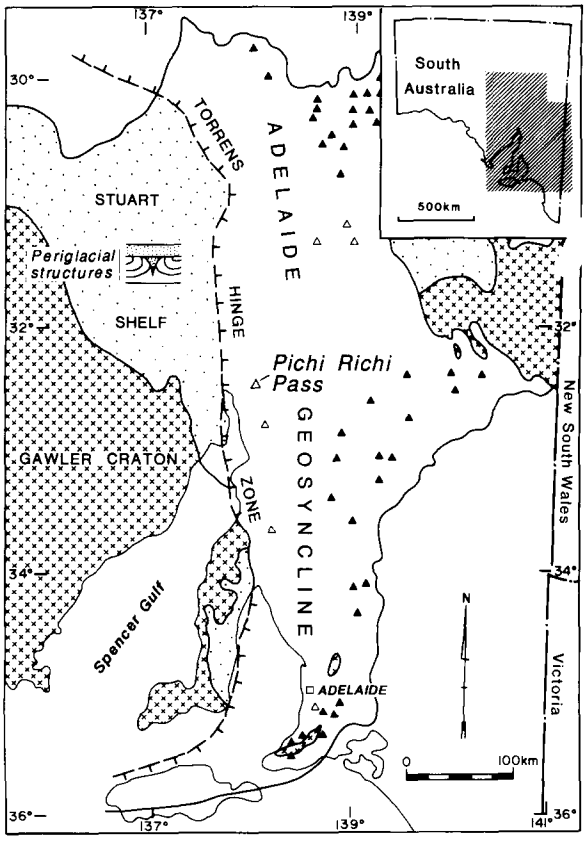


Fig. 1. Locality map of the Adelaide Geosyncline region, showing the distribution of glacial and periglacial facies of the Marinoan Glaciation. Solid triangles represent tillitic facies; open triangles, dropstone facies; dots, stable shelves; crosses, cratonic areas.

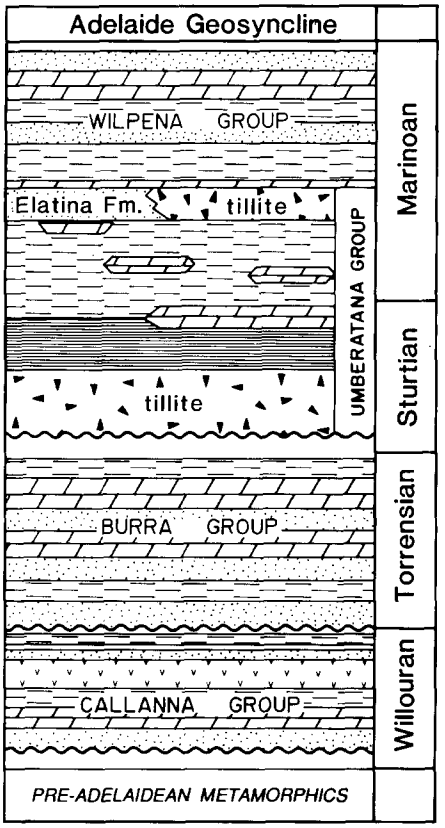


Fig. 2. Simplified stratigraphic column of the Late Proterozoic (Adelaidean) rocks in the Adelaide Geosyncline, showing the relationship of the Elatina Formation to tillitic facies. Adelaidean strata (aggregate thickness ~ 20 km) rest of metamorphics dated at ~ 850–1000 Ma, and are disconformably overlain by Early Cambrian deposits [1].

therefore be primary, dating closely to the time of deposition. The magnetisation has a low inclination (−9.6°) which is of great significance given the formation's glacial affinity.

More recently it has been reported [11] that the magnetisation of the rhythmite member pre-dates soft sediment deformation. This is further evidence that the remanence was acquired at or soon after deposition as a detrital remanent magnetisation (DRM). Irving and Major [12] suggested two types of DRM. The first, called depositional DRM, was thought to be acquired at the time of deposition and the second, called post-depositional DRM (p-DRM), was thought to be acquired sometime later. The essential difference between the two is that the direction of depositional DRM is deflected towards the bedding plane, producing an

inclination error [13,14]. It has been shown experimentally that post-depositional grain rotations enable some sediments to record accurately the ambient field direction [12]. Verosub [15] reviewed depositional and post-depositional magnetisation and arrived at the same conclusion. He speculated that post-depositional rotations are controlled by the nature and composition of both the magnetic carriers and the matrix and the pore water content at the time of deposition. Levi and Banerjee [16] found experimental evidence suggesting that inclination shallowing is minimised for fine-grained sediments.

The rotation of magnetic grains would conceivably be reflected in the magnetic fabric of a sediment and therefore the anisotropy of magnetic susceptibility (AMS) was suggested as a means of distinguishing between the magnetisation processes [15]. Other workers have determined experimental factors to correct for depositional deflection in both magnetite- [14] and haematite- bearing [17] sediments. More recently, Jackson et al. [18] suggested that the anisotropy of anhysteretic remanent magnetisation (ARM) susceptibility (AAS) may be used as a direct measure of the magnetisation deflection and to apply a correction.

The present study is aimed at determining the origin of magnetisation of the Elatina Formation to constrain further the probable palaeolatitude of the Late Proterozoic Marinoan Glaciation. We examine the timing of the remanence with respect to the soft sediment deformation and investigate ways to correct for any potential inclination error. We also address the possibility of the magnetisation recording a geomagnetic excursion or reversal. Furthermore, latitude-dependent palaeotidal



Fig. 3. Large rock sample from the tidal rhythmite member of the Elatina Formation at Pichi Richi Pass that was analysed in the present study. The scale is 3 cm long. Conspicuous dark mud drapes spaced ~ 0.5–1 cm apart bound 28 fortnightly lamina-cycles, and define a cuspate soft-sediment fold of probable gravity-slide origin; the top and base of the fold are not seen in the sample. The full size of the sample was 60 cm long × 16 cm stratigraphic thickness × 12 cm wide. In geographic coordinates the right-hand side of the sample was to the north and the joint face was dipping 73°E. The fold axis is oriented NW–SE.

data recorded by the rhythmite member are considered in the interpretation of the palaeomagnetic data.

2. Palaeomagnetic and rock magnetic methods

Routine palaeomagnetic methods [19–21] were used. Natural remanent magnetisation (NRM) and laboratory induced remanences were measured using either a CTF three-axis cryogenic magnetometer or an upgraded and modified fluxgate spinner magnetometer. The demagnetisers used were a Schonstedt AF demagnetiser model GSD-1 and the CSIRO three-stage carousel furnace. The furnace is housed inside a 4 m ten coil Helmholtz set with automatic feed-back maintaining zero-field ($\leqslant 5$ nT). The Schonstedt GSD-1 was used in conjunction with a 20 cm diameter $\times$ 50 cm long solenoid wound inside its mu-metal shield to induce anhysteretic remanent magnetisations (ARM). Magnetic susceptibility and anisotropy of magnetic susceptibility (AMS) were measured with the CSIRO transformer bridge [22] and an upgraded and modified DIGICO anisotropy delineator respectively. A water-cooled electromagnet was used to produce magnetic fields up to 1 T ($\sim$ 10,000 Oe) for (near) remanence saturation.

A single large, laminated rock sample (measuring 60 cm long $\times$ 16 cm stratigraphic thickness $\times$ 12 cm wide) showing soft-sediment deformation (Fig. 3) from the rhythmite member of the Elatina Formation was oriented using a magnetic compass/clinometer. The deformation, with a wavelength of 14–22 cm and an amplitude of up to 4 cm, appears to have resulted from low-angle gravity sliding. The angle between bedding plane poles varies by over 40°. Initially, the sample was intensively cored to yield 51 standard palaeomagnetic specimens. Careful alignment and marking of the specimens ensured that the between-specimen orientation is accurate to 1–2°. Apparent bedding planes within each specimen were visually enhanced with a marking pen and measured with a goniometer. Bedding plane poles for groups of similarly deformed specimens (Fig. 4) show a spread from the NW quadrant to the SE quadrant. The specimens were subjected to AF and thermal demagnetisation to isolate the most stable remanent component. A further 27 specimens were prepared later to study the anisotropy of magnetic susceptibility (AMS), the anisotropy of ARM susceptibility (AAS), hysteresis of remanence and the low-temperature behaviour of the remanence.

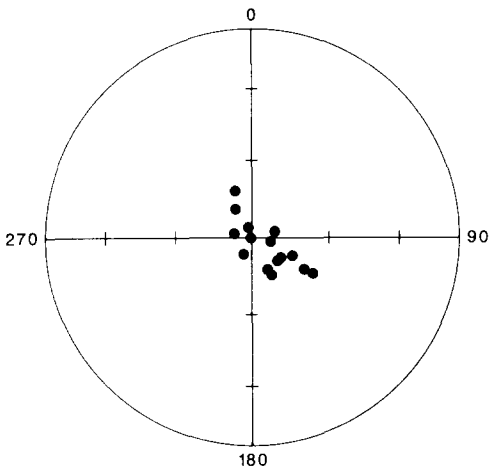


Fig. 4. Equal-area stereograph of average bedding plane poles for groups of specimens from the sample shown in Fig. 3.

3. Magnetic mineralogy

Mineralogical examination of a representative sample of opaque heavy mineral grains extracted from the Elatina Formation have shown that over 30% are haematite/martite grains, with exsolution blades and lamellae of rutile and ilmenite [5]. These grains suggest a high-temperature origin and may be confidently regarded as detrital. Other opaque grains such as ilmenite do not carry a magnetic remanence at normal temperatures and do not contribute to the NRM. Susceptibilities of $\sim$ 0.0001 SI (7–10 μG/Oe) and a Curie temperature of about 690°C attest to haematite being the main magnetic mineral present.

There is a controversy regarding the origin of remanence in haematite-bearing red beds. Some hold the view that haematite in sediments is predominantly an authigenic mineral, and that its remanence is a chemical remanent magnetisation (CRM) [23–25]. Others consider that haematite can acquire a stable detrital remanence at, or shortly after, the time of deposition (i.e. the remanence is a DRM) [26–28]. A third school believes that some red beds acquire a secondary CRM some time after they acquire their primary DRM [29,30]. In terms of the origin of the haematite, and therefore the remanence of the Elatina For-

mation, it is clearly important to establish the magnetic properties of the Elatina Formation for comparison with those of other red beds.

Hysteresis of remanence measurements provide additional information on the nature of the magnetic carrier. The remanences of specimens were measured after exposure to successively higher magnetic fields, yielding acquisition of remanence curves (Fig. 5a). The maximum magnetic field of 1 T (~ 10,000 Oe) available was not enough to produce saturation. However, by making the simple assumption that only the component of the field that is parallel to the easy axis of magnetisation of a grain is effective in switching its magnetisation, the equation for the remanence of an assemblage of non-interacting grains can be shown to be of the following form (equations 21 and 22 of [31]):

$$J_r = J_{rs}\left[1 - H_s^2/H^2\right]$$

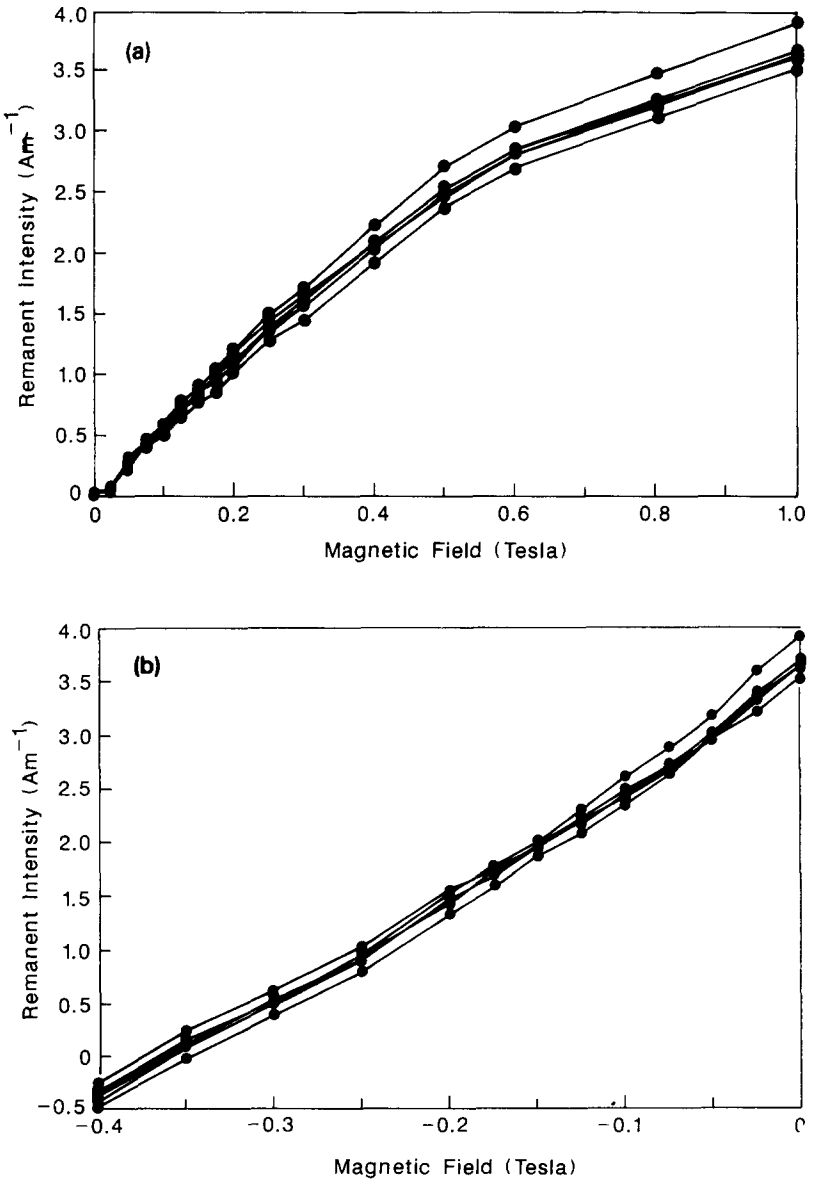


Fig. 5. Magnetic remanence hysteresis. (a) acquisition, and (b) demagnetisation (1A $m^{-1} = 10^3 \mu G = 10^3$ emu/cm^3, 1 T = 1000 Oe).

where J_r is the remanent magnetisation, J_{rs} is the saturation remanent magnetisation, H_s the switching field of the grains, and H the applied magnetic field. Saturation magnetisation may be estimated by extrapolating this trend to an infinite applied magnetic field. This yields a J_{rs} of about 4 A m^{-1} (4000 μG) and a H_c of about 0.3 T, or more strictly, 238 kA m^{-1} (3000 Oe). The switching field compares reasonably well with the coercivity of remanence of 286 kA m^{-1} (3600 Oe), as observed during DC demagnetisation after reversing the acquisition procedure (Fig. 5b). Such high coercivities point to haematite as the remanence carrier. It is noteworthy that Fig. 5 gives no indication of a second magnetic carrier. In particular, the presence of magnetite, which has a high saturation remanence and low coercivity, would be obvious in the hysteresis curves (cf. those presented by Dekkers and Linssen [32]). This supports the conclusion of the k–T analysis that haematite is the only remanence carrier.

4. Demagnetisation results and fold test

Remanence measurements following both the AF and thermal demagnetisation confirmed the exceptional stability of the remanence as found previously [5]. Figure 6 shows typical demagnetisation behaviour for the rhythmite samples of the Elatina Formation. The cleaned directions have been determined by linearity spectrum analysis (LSA [33]). Results from specimens have been combined in groups with similar bedding attitudes. This usually meant combining specimens from the same core although sometimes adjacent cores were also included. The mean results from these 15 groups are listed in Table 1. A comparison of the directions with respect to present horizontal and to their bedding planes is given in Fig. 7a and b. These directions have been rotated to the centre of the nets in Fig. 7c and d to expedite comparison. The tighter clustering on correcting for bedding is evident.

While it is clear that the magnetisation does not post-date the soft-sediment deformation, and may therefore be considered a DRM, the McFadden and Jones [34] test for a common mean direction is difficult to apply since the precisions (k) between groups vary too much. As can be determined from Table 1, Group 3 has a k of 3333,

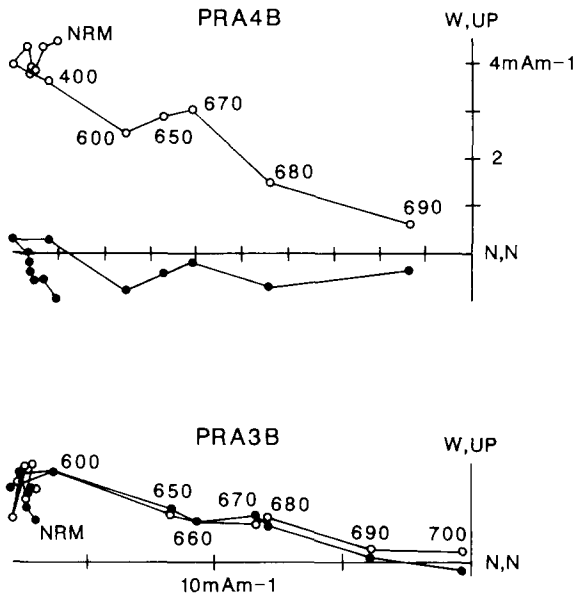


Fig. 6. Orthogonal plots showing thermal demagnetisation vectors for specimens PRA4B and PRA3B. Each plot consists of vector end-points (after heating and cooling in zero-field) projected onto two perpendicular planes that are superimposed with common N–S axes. Solid symbols refer to the horizontal plane (N–S and E–W) while open symbols refer to the vertical plane (N–S and up–down). Units are the same for horizontal and vertical axes, as indicated. Treatment steps shown are temperature in degrees centigrade.

while Group 4 has a k of 139. This wide variation reflects the small number of specimens in each group. If the wide variation is ignored and the test is applied regardless, the f-statistics calculated with respect to the present horizontal and the bedding plane are 13.07 and 5.20 respectively. The critical value of the F-distribution (with 28 and 72 degrees of freedom) is 2.01, suggesting that the groups do not have a common mean direction in either attitude. This is relevant to discussions below. However, the test has not been rigorously applied and since the precisions from some groups are so high, the errors introduced in estimating the bedding plane attitudes are probably significant. Whilst ignoring errors from this source does not affect the test with respect to present horizontal, it will affect testing for a common mean direction with respect to the bedding plane or for intermediate attitudes.

A more general test that is suitable for situations such as that described here makes use of bootstrap techniques [35]. Applying this test to the

data listed in Table 1 indicates that the tighter grouping after (complete) bedding correction should arise by chance with a probability of only 0.003, strongly suggesting that the magnetisation was acquired while the sediments were still in a plastic state. This test verifies the above conclusion based on visual inspection. These results are consistent with those of Sumner et al. [11].

A closer examination of the group mean magnetisation directions reveals that the most concentrated ($k = 475$) cluster actually occurs after 67% unfolding. The directions actually cross over each other on greater unfolding, explaining the result of the McFadden and Jones [34] test. This suggests that the magnetisation is syndeformational. An incremental "correlation" fold test recently developed is well adapted to this problem [38]. The method determines the degree of unfolding that minimises the correlation between the down dip azimuth and the azimuth between the site (group) mean direction and overall mean direction (after suitably transforming the coordinates such that the overall mean is vertically directed). Applying this reveals that a minimum correlation, and therefore a maximum probability of remanence acquisition, occurs with the beds 67% unfolded. The corresponding group mean directions and overall mean direction are listed in Table 1. The tighter cluster of directions after 67% unfolding, compared with the *in situ* and completely unfolded clusters (Fig. 7a–d), are shown in Fig. 7e and f. This result slightly modifies the findings of Sumner et al. [11].

Two interpretations of syndeformational magnetisations have been proposed. The cross-over may simply imply that the magnetisation was acquired when the beds were in some intermediate stage of deformation [37–40]. The other possibility is that a pre-existing remanence has been rotated during simple shear strain during deformation [41–43]. In terms of the timing of the magnetisation of the Elatina Formation, which interpretation is correct is not important because the plastic deformation occurred soon after deposition, indi-

TABLE 1

Summary of palaeomagnetic results for soft-sediment folds in the Elatina rhythmites, before and after unfolding

Group	Bedding poles				Remanence					
					Before		After 67% corr.		After 100% corr.	
	N	R	Az (°)	Pl (°)	Dec (°)	Inc (°)	Dec (°)	Inc (°)	Dec (°)	Inc (°)
1	3	2.997	164.9	81.4	179.3	−6.9	179.2	−1.8	179.2	1.4
2	3	2.996	217.8	76.7	183.6	−10.3	184.2	−3.6	184.2	0.8
3	3	2.999	24.4	89.0	189.0	−6.8	189.0	−7.4	189.0	−6.8
4	3	2.986	209.6	63.0	179.5	−22.8	181.7	−8.4	182.1	1.0
5	3	2.993	211.2	66.6	183.5	−22.8	185.4	−10.0	185.8	−1.8
6	4	3.998	204.5	73.1	181.2	−17.9	182.1	−8.4	182.4	−2.3
7	3	2.997	241.2	73.8	188.8	−9.6	189.7	−3.5	189.8	0.4
8	3	2.997	244.9	76.3	186.9	−10.7	187.9	−6.2	188.3	−3.3
9	3	2.998	300.5	83.1	184.5	−3.3	184.8	−5.1	185.0	−6.3
10	4	3.994	221.2	77.1	188.0	−12.4	188.7	−5.8	188.9	−1.5
11	4	3.994	191.5	82.7	185.5	−10.3	185.6	−5.9	185.6	−3.0
12	3	2.992	8.8	83.1	185.8	−3.7	185.8	−7.9	185.8	−10.6
13	4	3.992	65.8	86.2	186.3	−0.6	186.3	−1.8	186.2	−2.5
14	5	4.985	58.6	77.6	184.5	−1.2	184.1	−5.6	183.6	−8.4
15	3	2.997	69.2	70.9	184.3	−1.8	183.5	−6.7	182.5	−9.7
Mean 15					184.7	−9.4	185.2	−5.9	185.2	−3.6
					$R = 14.871$		$R = 14.969$		$R = 14.947$	
					$k = 108$		$k = 461$		$k = 262$	
					$\alpha_{95} = 3.7°$		$\alpha_{95} = 1.8°$		$\alpha_{95} = 2.4°$	

N = number of specimens; R = resultant length of unit vectors; k = precision parameter; α_{95} = half-angle of cone of 95% confidence [54].

cating clearly that the remanence was acquired very soon after the formation was deposited. Although in other situations we have had reasons to favour the remagnetisation model of the cross-over signature [39,40], given the plastic nature of the Elatina Formation at the time of its deformation we feel that the deformational model may apply here. Situations probably exist where each interpretation has currency, depending upon the rock type and deformation style.

Assuming that the deformational, or shear strain model of syndeformational magnetisation produces symmetrical rotation of remanence vectors, there should be no bias in the overall mean direction if sampling is uniformly distributed. The present study satisfies this condition, so estimation of the palaeolatitude of deposition of the Elatina Formation should not be biased by syndeformational effects. However, it is important to the question of palaeolatitude whether the remanence is a depositional DRM or a p-DRM and therefore whether an inclination error [14] may exist. In this regard it is noteworthy that the fold structure (Fig. 3) includes at least 28 fortnightly lamina-cycles, indicating that sediments representing at least 1.1 yr of deposition are involved. As the very fine sands and silts comprising the Elatina rhythmites were deposited mainly from

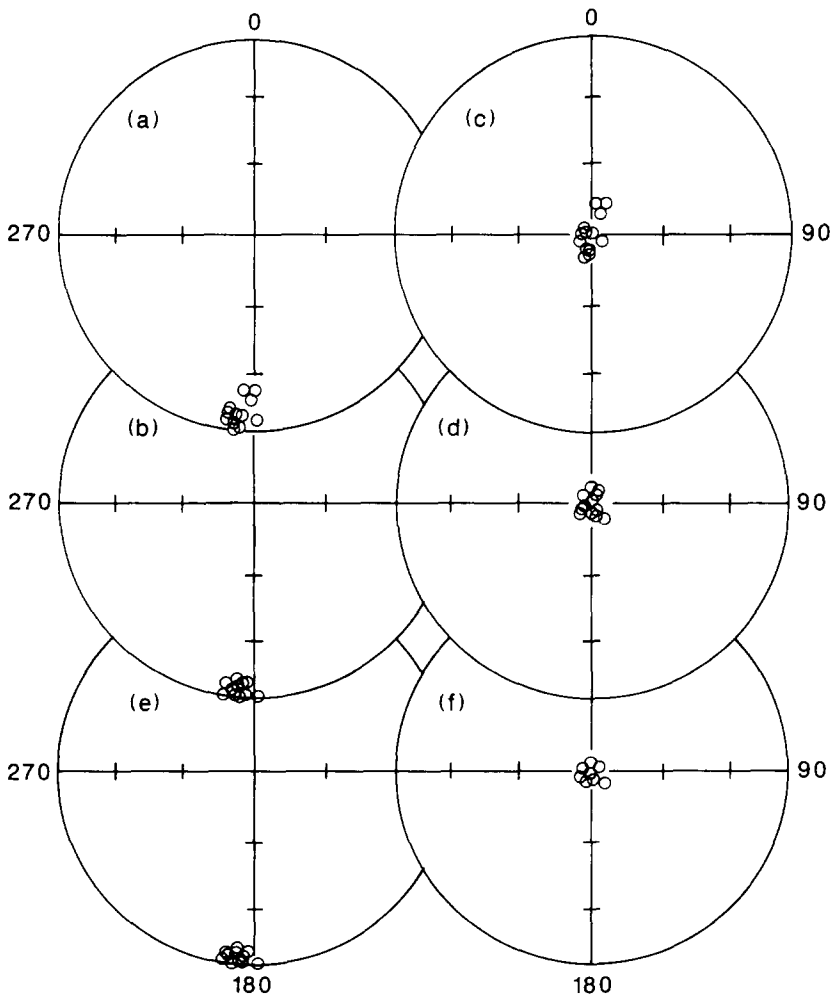


Fig. 7. Equal-area stereographs of cleaned mean directions of natural remanent magnetisation (NRM). (a) *In situ*; (b) completely unfolded; (c) and (d) the previous sets of directions rotated to the vertical to facilitate comparison; (e) 67% unfolded; and (f) the set of directions in (e) rotated to the vertical.

suspension, they would likely have undergone some dewatering and rotation of particles during the interval of 1.1 + yr prior to the formation of the fold. Hence, the magnetisation of the rhythmites may well have been acquired (or modified) during the 1.1 + yr that the sediment lay on the sea floor before sliding downslope. Experiments show that p-DRM can be acquired in less than one year in a fine sediment deposited from suspension [15].

5. Inclination error

Since an inclination error would bias the calculated palaeolatitude equatorward (assuming an axial geocentric dipole model), it is important to quantify any such deflection. This may be attempted using the empirical relationship of King [14],

$$\tan(I_o) = f \tan(I_f)$$

where I_o and I_f are the observed remanence inclination and the magnetic field inclination respectively, and f is a constant of proportionality. Values for f have been found experimentally to vary from 0.55 for haematite-bearing sediments [17] to 0.4 for magnetite-bearing sediments [14]. Curiously, the relative values of f are the inverse of that expected, given the much greater magnetic coupling of magnetite. However, assuming a value of 0.5, and depositional DRM, the mean inclination of the palaeomagnetic direction with respect to bedding increases (in magnitude) from −5.9 to −11.7°. However, Tauxe and Kent [17] caution against the uncritical use of this procedure, noting that if p-DRM is also present the procedure will over-correct. The −11.7° is therefore a maximum value.

Another method to correct for possible inclination error has been proposed recently. Jackson et al. [18] suggest that the anisotropy of anhysteretic remanent magnetisation susceptibility (AAS) tensor relates the remanence direction to the magnetic field direction. However, since a maximum alternating field of 350 mT (3500 Oe) was available, only a fraction of the haematite grains can be activated, or switched. Anisotropy of isothermal remanent magnetisation (IRM) using DC fields of up to 1000 mT (10 kOe) seems to offer an alternative but it has been shown recently [44] that exposure of haematite to high fields changes its coercivity. The DC method also presents the problem of demagnetising the remanence between successive IRM acquisitions.

A reliable method of estimating any inclination error therefore is not available. However, as discussed above, a p-DRM origin for the magnetisation seems more likely, which implies that an inclination error is implausible. The good agreement between the Elatina pole position and other Late Proterozoic poles from different rock units of the Adelaide Geosyncline (Fig. 8 and Table 2) further argues that the magnetisation of the Elatina Formation is not significantly affected by an in-

TABLE 2

Palaeomagnetic data for Late Proterozoic rocks, South Australia

Pole	Rock unit	Age (Ma)	Palaeomagnetic pole				Ref.
			Lat. (°S)	Long. (°E)	K	A_{95} (°) dp,dm (°)	
AC	Acraman meltrock	~ 600	8.6	353.4	78.3	6,3	[55]
BU	Bunyeroo Fm.	~ 600	7.0	17.0	19.8	9,14	[56]
BRM	Brachina Fm.	~ 600–650	33	328	–	12,20	[56]
BRK	Brachina Fm.	~ 600–650	45	334	–	4,7	[56]
EF	Elatina Fm.	~ 650	50.9	337.1	–	2,2	[5]
AF	Angepena Fm.	~ 700	33	344	–	9,16	[56]
TH	Tapley Hill Fm.	750 ± 53	39	28	–	8,15	[56]
MT	Merinjina Tillite	~ 750–800	45	346	–	6,11	[56]
CQ	Copley Quartzite	~ 800	36	10	–	7,15	[56]

K = precision parameter; A_{95} = half-angle of cone of 95% confidence for poles [56]; dp,dm = semi-minor, semi-major axes polar error ellipse. All rock units except Acraman meltrock are from the Adelaide Geosyncline.

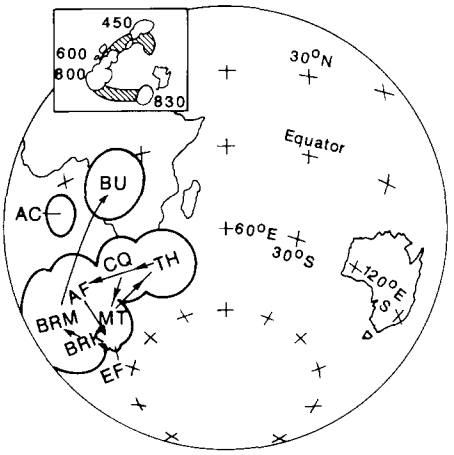


Fig. 8. Late Proterozoic to Early Palaeozoic apparent polar wander path for the Adelaide Geosyncline. See Table 2 for explanation of Late Proterozoic poles.

clination error. As discussed in the following section, the same good agreement indicates that the magnetisation does not record a geomagnetic excursion or reversal.

6. The Elatina palaeopole and palaeolatitude of deposition

Since the large rock sample studied here (Fig. 3) spans only ~ 1.1 yr of deposition, the indicated pole (Table 1) must be considered a virtual geomagnetic pole (VGP) that represents an instantaneous sample of the geomagnetic field, which is generally subject to secular variation. The pole previously determined for the Elatina rhythmites by Embleton and Williams [5] likewise must be viewed as a VGP because, by the tidal interpretation of the rhythmites [7–10], the full rhythmite member records only ~ 60–70 yr deposition. Attributing all the palaeomagnetic data to a single VGP accounts for the previously observed low directional dispersion of the remanence [5].

The similarity of the Elatina VGP to palaeomagnetic poles for other Late Proterozoic rocks in South Australia (Fig. 8 and Table 2) argues strongly, however, that the Elatina VGP does not record an excursion or reversal of the geomagnetic field. Pole positions from units both stratigraphically below and above the Elatina Formation are similar (Table 2), and the possibility of remagnetisation is eliminated by positive fold tests for both the Elatina Formation and the Bunyeroo Formation. Indeed, the Elatina palaeomagnetic data satisfy six of the seven reliability criteria of Van der Voo [45], the only criterion not fulfilled being of course the presence of reversals. The Elatina data are rare amongst Precambrian palaeomagnetic results in being constrained by a field test.

Palaeomagnetic latitudes indicated for the Elatina rhythmite member are $3.0° \pm 1.8°$N (present study, 67% unfolded data) and $4.8° \pm 2.2°$N [5]. As these values are relative to a VGP, the amplitude of secular variation should be considered to estimate the limits of the true palaeolatitude of deposition. Little is known about secular variation during the Proterozoic although compilations of data for the Phanerozoic show that equatorial values of the circular standard deviation have generally been between 14 and 18° [46]. We may conclude, therefore, that the palaeolatitude of deposition of the Elatina Formation was $4 \pm 16°$N. This conclusion is consistent with sedimentological evidence for a clear diurnal inequality in the palaeotides during deposition of the Elatina Formation and its equivalent, the Reynella Siltstone [7–10,47]. The diurnal inequality of the tides is minimal or absent at the equator [48] but can be conspicuous at latitudes of 13–14° [49,50], which suggests that the Late Proterozoic rhythmites were deposited in low rather than equatorial palaeolatitudes. The present results are in good agreement with the depositional palaeolatitude of about 12°N for the Elatina Formation indicated by the mean palaeomagnetic pole for Australia for the interval 700–560 Ma [5].

7. Discussion and conclusions

This study has confirmed the findings of Embleton and Williams [5] showing that the magnetisation of the Elatina Formation rhythmite member is very stable and was acquired at or soon after deposition. The direction of remanence is different from known directions for Australian Phanerozoic rocks but is similar to those found in other Late Proterozoic rocks in the Adelaide Geosyncline. The remanence is most probably carried by detrital haematite of high-temperature origin, and because the sediments have undergone only burial diagenesis this suggests that the magnetic

carriers have not been substantially altered since deposition. The execution of a fold test on an apparent gravity-slide structure suggests that most of the remanence of the Elatina Formation was acquired prior to the soft-sediment deformation.

Although it is not possible to distinguish conclusively between a model of remanence acquisition involving a partial depositional/post-depositional DRM, and a model involving syndeformational magnetisation, the remanence is carried by haematite of high-temperature origin suggesting magnetisation was acquired early. The magnetisation is characterised by a very stable single component, with no secondary overprinting. All evidence points to the Elatina Formation acquiring most its remanence during or very soon after deposition, with slight modification during soft-sediment deformation.

The Elatina pole must be considered a virtual geomagnetic pole, because the tidal rhythmite member studied represents only ~ 60–70 yr deposition. The similarity of the Elatina VGP to palaeomagnetic poles for other Late Proterozoic rocks in South Australia provides strong evidence, however, that the Elatina VGP does not record an excursion or reversal of the geomagnetic field and that the magnetisation direction has not been significantly affected by inclination error.

The low latitude ($< 10°$) therefore implies a palaeolatitude of deposition within the approximate limits of 20°N to 12°S of the palaeoequator. As the Elatina Formation is part of the Late Proterozoic (~ 650 Ma) Marinoan glacial succession in South Australia, the results of the present study therefore provide strong evidence for a low palaeolatitude of Late Proterozoic glaciation.

As discussed by Williams [51] and Embleton and Williams [5], three principal hypotheses have been advanced to explain the palaeomagnetic evidence from several continents for a preferred low palaeolatitude ($< 30°$) of glaciation and *in situ* frigid climate near sea level during the Late Proterozoic (~ 800–600 Ma):

(1) Glaciation occurred from pole to pole during very severe ice ages.

(2) The axial geocentric dipole model for the Earth's magnetic field is invalid, at least for that time interval.

(3) A reversed climatic zonation and marked seasonality prevailed because of an increased obliquity of the ecliptic ($> 54°$) in Late Proterozoic time.

A lack of continents in high palaeolatitudes during global glaciation might explain the apparent absence of high-palaeolatitude glaciation between ~ 800 and 600 Ma [52]. However, the strong seasonality with regard to temperature in low palaeolatitudes indicated by seasonal freeze–thaw structures in coeval fossil permafrost horizons (mean monthly temperature ranges as great as $< -35°C$ in midwinter to $+4°C$ in midsummer [3,4]) militates against the global-glaciation hypothesis [5,51]. Discrimination between the remaining two hypotheses requires palaeogeophysical data that are independent of both palaeoclimate and palaeomagnetism. It may be significant, therefore, that the palaeotidal data for Late Proterozoic time recorded by the tidal rhythmites of the Elatina Formation [7–10,47] are consistent with both a low palaeolatitude of deposition and a substantial obliquity of the ecliptic at that time [53].

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